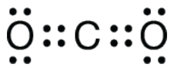
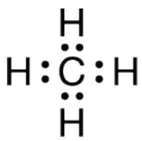
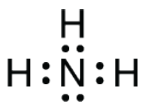
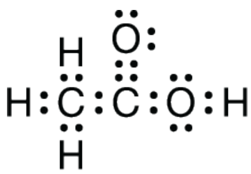
Chapter 02 Part 2: Molecular Structures

1. Which of the following complete Lewis diagrams represents a molecule containing a bond angle that is closest to 120°?

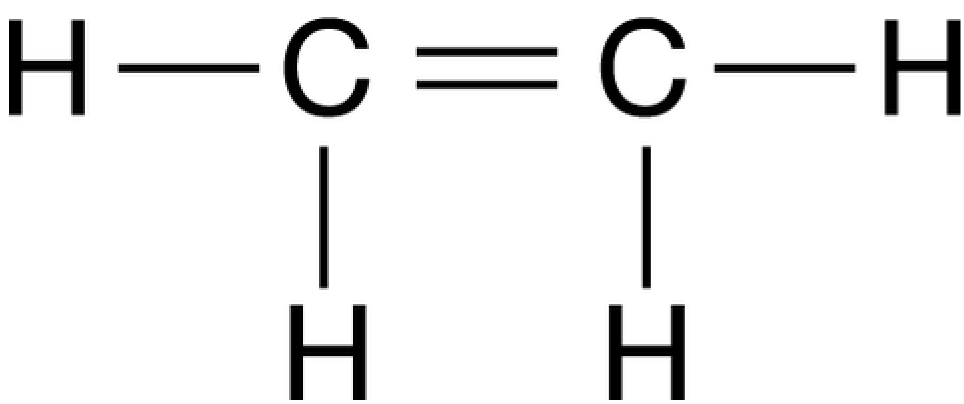
(A) 

(B) 

(C) 

(D) 

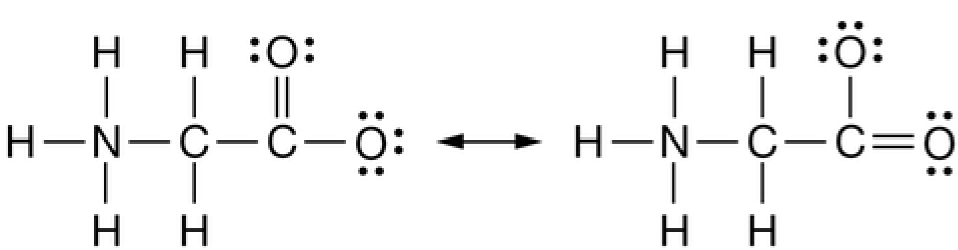
2.



A Lewis diagram for the molecule C_2H_4 is shown above. In the actual C_2H_4 molecule, the H-C-H bond angles are closest to

(A) 90°
(B) 109.5°
(C) 120°
(D) 180°

3.



Based on the resonance structures shown above, what are the bond orders of the two carbon-oxygen bonds?

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(A) 1 and 1.5
(B) 1 and 2
(C) 1.5 and 1.5
(D) 2 and 2

Reaction 1: $CaC_2(s) + 2\ H_2O(l) \rightarrow C_2H_2(g) + Ca(OH)_2(s)$

Reaction 2: $NaOCl(aq) + 2\ HCl(aq) \rightarrow Cl_2(g) + NaCl(aq) + H_2O(l)$

Reaction 3: $C_2H_2(g) + Cl_2(g) \rightarrow C_2H_2Cl_2(g)$

4. When Reaction 3 occurs, does the hybridization of the carbon atoms change?

(A) Yes; it changes from *sp* to sp^2.
(B) Yes; it changes from *sp* to sp^3.
(C) Yes; it changes from sp^2 to sp^3.
(D) No; it does not change

5.

Lewis electron-dot diagrams for CO_2 and SO_2 are given above. The molecular geometry and polarity of the two substances are

(A) the same because the molecular formulas are similar
(B) the same because C and S have similar electronegativity values
(C) different because the lone pair of electrons on the S atom make it the negative end of a dipole
(D) different because S has a greater number of electron domains (regions of electron density) surrounding it than C has

---

Refer to the following diatomic species

(A) Li_2

(B) B_2

(C) N_2

(D) O_2

(E) F_2

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6. Contains 1 sigma (σ) and 2 pi (π) bonds

(A) Li_2
(B) B_2
(C) N_2
(D) O_2
(E) F_2

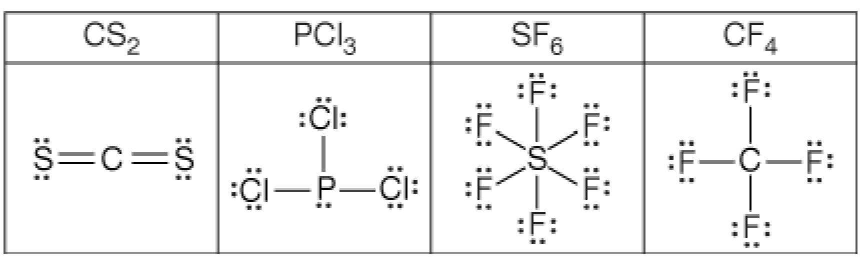
7. Has a bond order of 2

(A) Li_2
(B) B_2
(C) N_2
(D) O_2
(E) F_2

8. Has the largest bond-dissociation energy

(A) Li_2
(B) B_2
(C) N_2
(D) O_2
(E) F_2

9.



Based on the diagrams shown above, which of the following would have the greatest dipole moment, and why?

(A) CS_2, because the double bonds create an area of higher electron density around the C atom.
(B) PCl_3, because the Cl atoms draw electron density away from the P atom and the bond dipoles do not cancel one another.
(C) SF_6, because it has the greatest number of polar bonds.
(D) CF_4, because the large electronegativity difference between C and F results in very polar bonds.

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Directions: Each set of lettered choices below refers to the numbered statement immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

Refer to the following gaseous molecules:

(A) $BeCl_2$

(B) SO_2

(C) N_2

(D) O_2

(E) F_2

10. Is best represented by two or more resonance forms

(A) $BeCl_2$
(B) SO_2
(C) N_2
(D) O_2
(E) F_2

11. Is the molecule in which the intramolecular forces are strongest

(A) $BeCl_2$
(B) SO_2
(C) N_2
(D) O_2
(E) F_2

12. Is a polar molecule

(A) $BeCl_2$
(B) SO_2
(C) N_2
(D) O_2
(E) F_2

---

13. The geometry of the SO_3 molecule is best described as

(A) trigonal planar

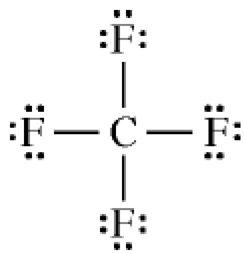
(B) trigonal pyramidal

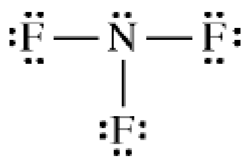
(C) square pyramidal

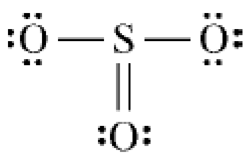
(D) bent

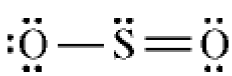
(E) tetrahedral

14. Which of the following Lewis electron-dot diagrams represents the molecule that contains the smallest bond angle?

(A) 

(B) 

(C) 

(D) 

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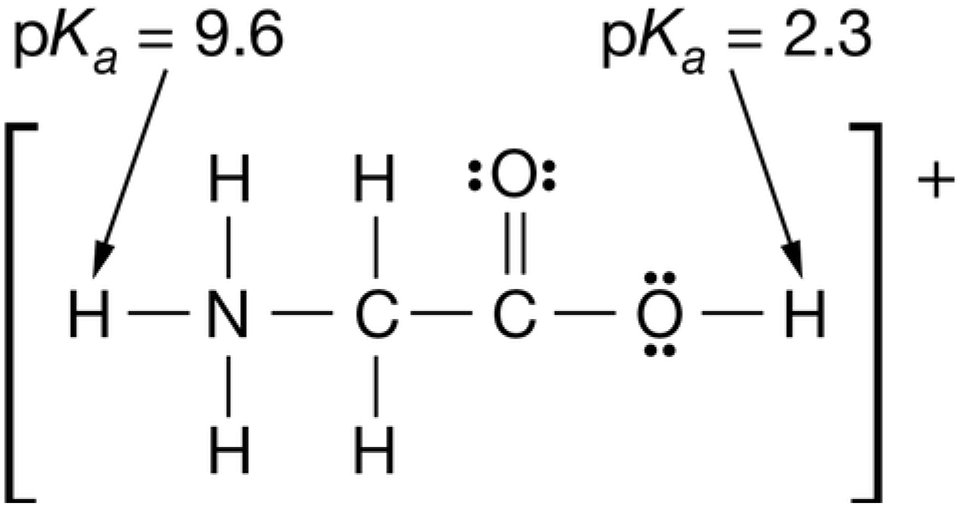

The structural formula of the glycinium cation is shown above. Arrows indicate the $\mathrm{p}K_a$ values for the labile protons in the molecule.

15. Which of the following is true about the geometry of the glycinium cation?

(A) The leftmost C atom and all the atoms directly bonded to it lie in the same plane.

(B) Both C atoms and both O atoms lie in the same plane.

(C) The $\mathrm{N-C-C}$ bond angle is 180°.

(D) The geometry around the N atom is planar.

16. What is the approximate $\mathrm{H-O-C}$ bond angle in the glycinium cation?

(A) 180°

(B) 120°

(C) 105°

(D) 90°

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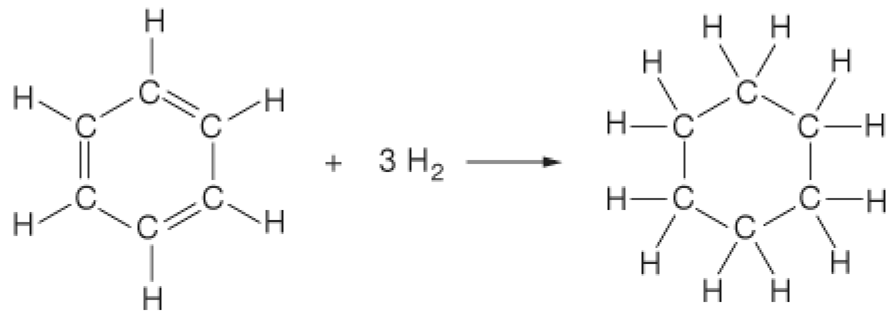
17.

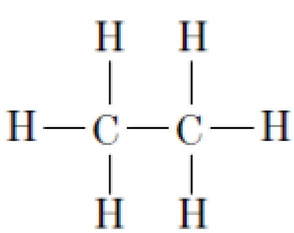


In the reaction represented above, which of the following sets correctly identifies the hybridization of the C atoms before and after the reaction occurs?

Set	**Before**	**After**
A	sp^2	sp^2
B	sp^2	sp^3
C	sp^3	sp^2
D	sp^3	sp^3

(A) Set A
(B) Set B
(C) Set C
(D) Set D

18.



The hybridization of the carbon atoms in the molecule represented above can be described as

(A) sp
(B) sp^2
(C) sp^3
(D) dsp^2
(E) d^2sp

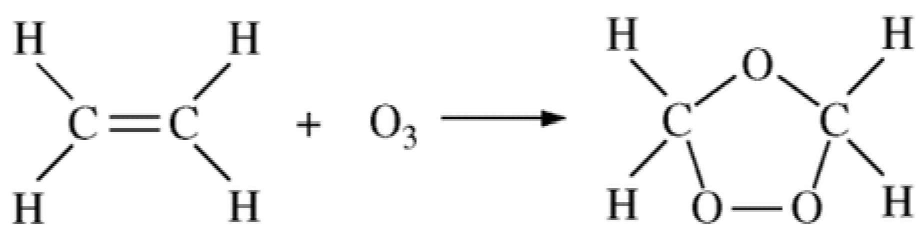
19.

$$H - C \equiv C - H$$

What is the hybridization of the carbon atoms in a molecule of ethyne, represented above?

(A) sp
(B) sp^2
(C) sp^3
(D) dsp^2
(E) d^2sp

20.



In the reaction represented above, what is the hybridization of the C atoms before and after the reaction occurs?

(A)

Before	After
sp	sp^2

(B)

Before	After
sp	sp^3

(C)

Before	After
sp^2	sp

(D)

Before	After
sp^2	sp^3

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

The question refers to the following species.

(A) H_2O
(B) NH_3
(C) BH_3
(D) CH_4
(E) SiH_4

21. Has a central atom with less than an octet of electrons

(A) H_2O
(B) NH_3
(C) BH_3
(D) CH_4
(E) SiH_4

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22. Is predicted to have the largest bond angle

(A) H_2O
(B) NH_3
(C) BH_3
(D) CH_4
(E) SiH_4

23. Has a trigonal-pyramidal molecular geometry

(A) H_2O
(B) NH_3
(C) BH_3
(D) CH_4
(E) SiH_4

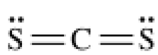
24. Has two lone pairs of electrons

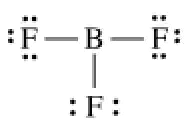
(A) H_2O
(B) NH_3
(C) BH_3
(D) CH_4
(E) SiH_4

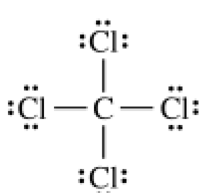
---

25. Which of the following molecules has the largest dipole moment?

(A) 

(B) 

(C) 

(D) 

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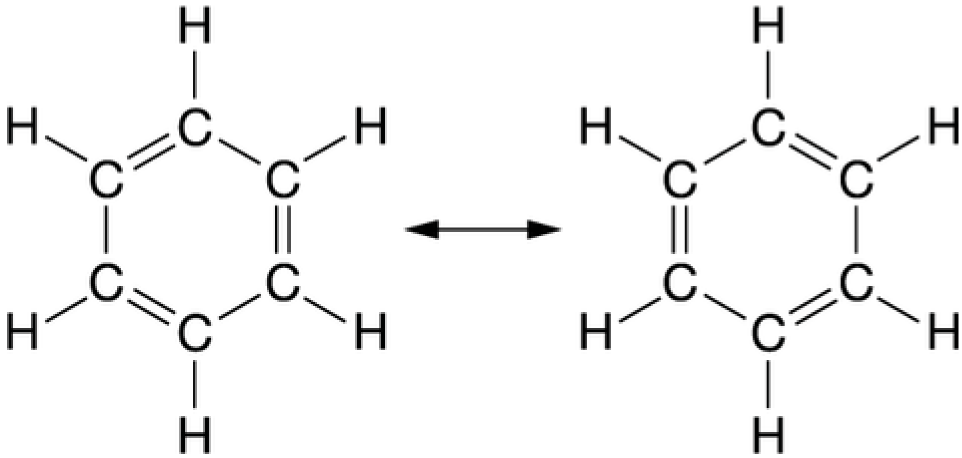
26.



The diagram above shows two resonance structures for a molecule of C_6H_6. The phenomenon shown in the diagram best supports which of the following claims about the bonding in C_6H_6 ?

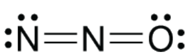
(A) In the C_6H_6 molecule, all the bonds between the carbon atoms have the same length.

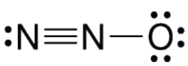
(B) Because of variable bonding between its carbon atoms, C_6H_6 is a good conductor of electricity.

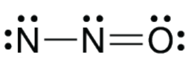
(C) The bonds between carbon atoms in C_6H_6 are unstable, and the compound decomposes quickly.

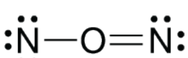
(D) The C_6H_6 molecule contains three single bonds between carbon atoms and three double bonds between carbon atoms.

27. Which of the following Lewis diagrams best represents the bonding in the N_2O molecule, considering formal charges?

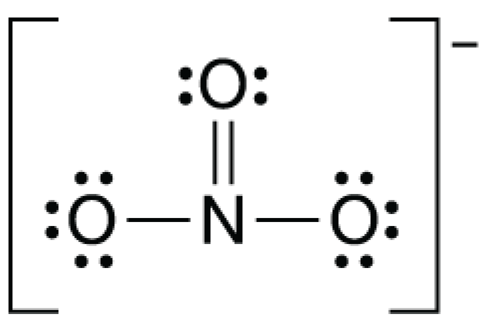
(A) 

(B) 

(C) 

(D) 

28.



Which of the following statements, if true, would support the claim that the $NO_3{}^-$ ion, represented above, has three resonance structures?

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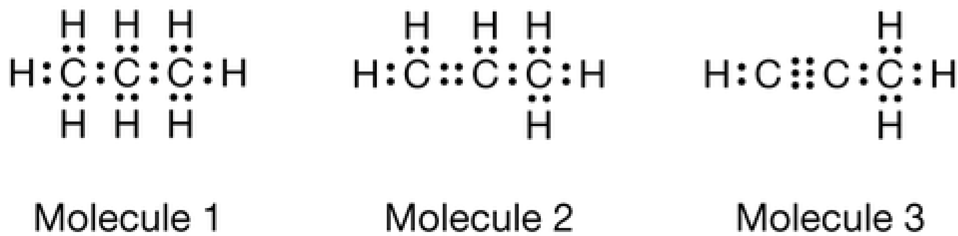
(A) The NO_3^- ion is not a polar species.

(B) The oxygen-to-nitrogen-to-oxygen bond angles are 90°.

(C) One of the bonds in NO_3^- is longer than the other two.

(D) One of the bonds in NO_3^- is shorter than the other two.

29.



Lewis diagrams of molecules of three different hydrocarbons are shown above. Which of the following claims about the molecules is best supported by the diagrams?

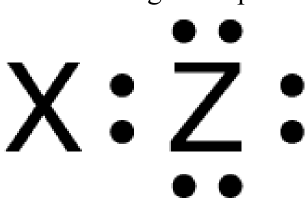
(A) All the atoms in molecule 1 lie in one plane.

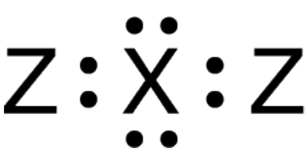
(B) All the molecules have the same empirical formula.

(C) The C-C-C bond angle in molecule 2 is close to 180°.

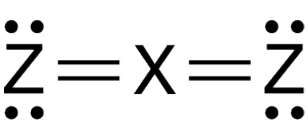
(D) The strongest carbon-to-carbon bond occurs in molecule 3.

30. In the following diagrams, elements are represented by X and Z, which form molecular compounds with one another. Which diagram represents a molecule that has a bent molecular geometry?

(A) 

(B) 

(C) 

(D) 

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Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

The following questions refer to the below molecules:

(A) CO_2
(B) H_2O
(C) CH_4
(D) C_2H_4
(E) PH_3

31. The molecule with the largest dipole moment

(A) CO_2
(B) H_2O
(C) CH_4
(D) C_2H_4
(E) PH_3

32. The molecule with only one double bond

(A) CO_2
(B) H_2O
(C) CH_4
(D) C_2H_4
(E) PH_3

33. The molecule that has trigonal pyramidal geometry

(A) CO_2
(B) H_2O
(C) CH_4
(D) C_2H_4
(E) PH_3

---

34. For which of the following molecules are resonance structures necessary to describe the bonding satisfactorily?

(A) H_2S
(B) SO_2
(C) CO_2
(D) OF_2
(E) PF_3

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35. Which of the following molecules contains exactly three sigma (σ) bonds and two pi (π) bonds?

(A) C_2H_2
(B) CO_2
(C) HCN
(D) SO_3
(E) N_2

36. Which of the following molecules has an angular (bent) geometry that is commonly represented as a resonance hybrid of two or more electron-dot structures?

(A) CO_2
(B) O_3
(C) CH_4
(D) BeF_2
(E) OF_2

37. Of the following molecules, which has the largest dipole moment?

(A) CO
(B) CO_2
(C) O_2
(D) HF
(E) F_2

38. Has molecules with a pyramidal shape

(A) $NH_3(g)$
(B) $BH_3(g)$
(C) $H_2(g)$
(D) $H_2S(g)$
(E) HBr(g)

39. Which of the following Lewis electron-dot diagrams represents the molecule that is the most polar?

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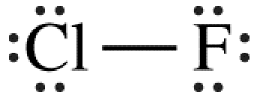
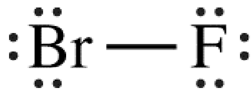
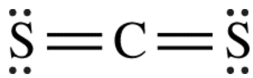
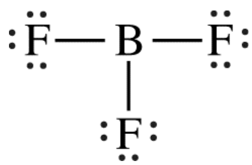
(A) 

(B) 

(C) 

(D) 

40. Which of the following molecules contains polar covalent bonds but is a nonpolar molecule?

(A) CH_3Cl
(B) CH_2Cl_2
(C) NH_3
(D) CCl_4
(E) N_2

41. Which of the following molecules is nonpolar but has polar covalent bonds?

(A) N_2
(B) H_2O_2
(C) H_2O
(D) CCl_4
(E) CH_2Cl_2

42. Which of the following arranges the molecules N_2, O_2, and F_2 in order of their bond enthalpies, from least to greatest?

(A) $F_2 < O_2 < N_2$
(B) $O_2 < N_2 < F_2$
(C) $N_2 < O_2 < F_2$
(D) $N_2 < F_2 < O_2$

43. Pi (π) bonding occurs in each of the following species EXCEPT

(A) CO_2
(B) C_2H_4
(C) CN^-
(D) C_6H_6
(E) CH_4

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44. Resonance is most commonly used to describe the bonding in molecules of which of the following?

(A) CO_2
(B) O_3
(C) H_2O
(D) CH_4
(E) SF_6

45. Which of the following molecules has the shortest bond length?

(A) N_2
(B) O_2
(C) Cl_2
(D) Br_2
(E) I_2

46. NH_3 reacts with BF_3 to form a single species. Which of the following structural diagrams is the most likely representation of the product of the reaction?

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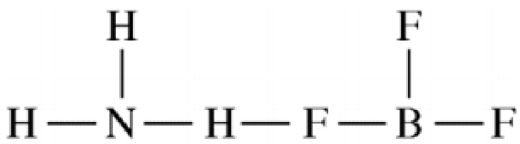
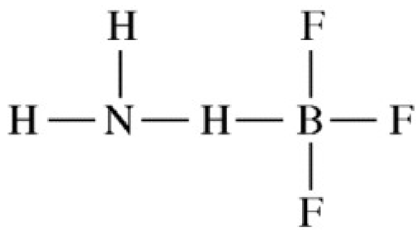
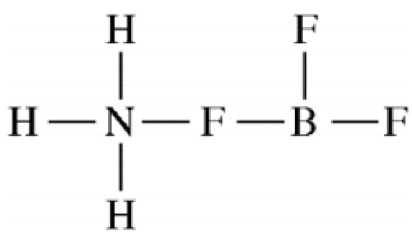
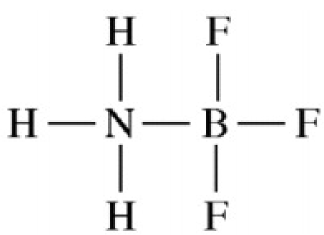
(A) 

(B) 

(C) 

(D) 

47. According to the VSEPR model, the progressive decrease in the bond angles in the series of molecules CH_4, NH_3, and H_2O is best accounted for by the

(A) increasing strength of the bonds
(B) decreasing size of the central atom
(C) increasing electronegativity of the central atom
(D) increasing number of unshared pairs of electrons
(E) decreasing repulsion between hydrogen atoms

48. The electron-dot structure (Lewis structure) for which of the following molecules would have two unshared pairs of electrons on the central atom?

(A) H_2S
(B) NH_3
(C) CH_4
(D) HCN
(E) CO_2

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49. Which of the following is a nonpolar molecule that contains polar bonds?

(A) F_2
(B) CHF_3
(C) CO_2
(D) HCl
(E) NH_3

50. Which of the molecules represented below contains carbon with *sp*^{2} hybridization?

(A) CH_4
(B) CH_2Cl_2
(C) C_2H_6
(D) $C_2H_2Cl_2$
(E) $C_2H_4Cl_2$

51. Which of the following molecules contains only single bonds?

(A) CH_3COOH
(B) $CH_3CH_2COOCH_3$
(C) C_2H_6
(D) C_6H_6
(E) HCN

52. Which of the following has a zero dipole moment?

(A) HCN
(B) NH_3
(C) SO_2
(D) NO_2
(E) PF_5

53. Which of the following species is NOT planar?

(A) CO_3^{2-}
(B) NO_3^-
(C) ClF_3
(D) BF_3
(E) PCl_3

54. The BF_3 molecule is nonpolar, whereas the NF_3 molecule is polar. Which of the following statements accounts for the difference in polarity of the two molecules?

(A) In NF_3, each F is joined to N with multiple bonds, whereas in BF_3 , each F is joined to B with single bonds.
(B) N — F bonds are polar, whereas B — F bonds are nonpolar.
(C) NF_3 is an ionic compound, whereas BF_3 is a molecular compound.
(D) Unlike BF_3, NF_3 has a nonplanar geometry due to an unshared pair of electrons on the N atom.

55. Account for each of the following observations in terms of atomic theory and/or quantum theory.

a. Atomic size decreases from Na to Cl in the periodic table.

b. Boron commonly forms molecules of the type BX_3. These molecules have a trigonal planar structure.

c. The first ionization energy of K is less than that of Na.

d. Each element displays a unique gas-phase emission spectrum.

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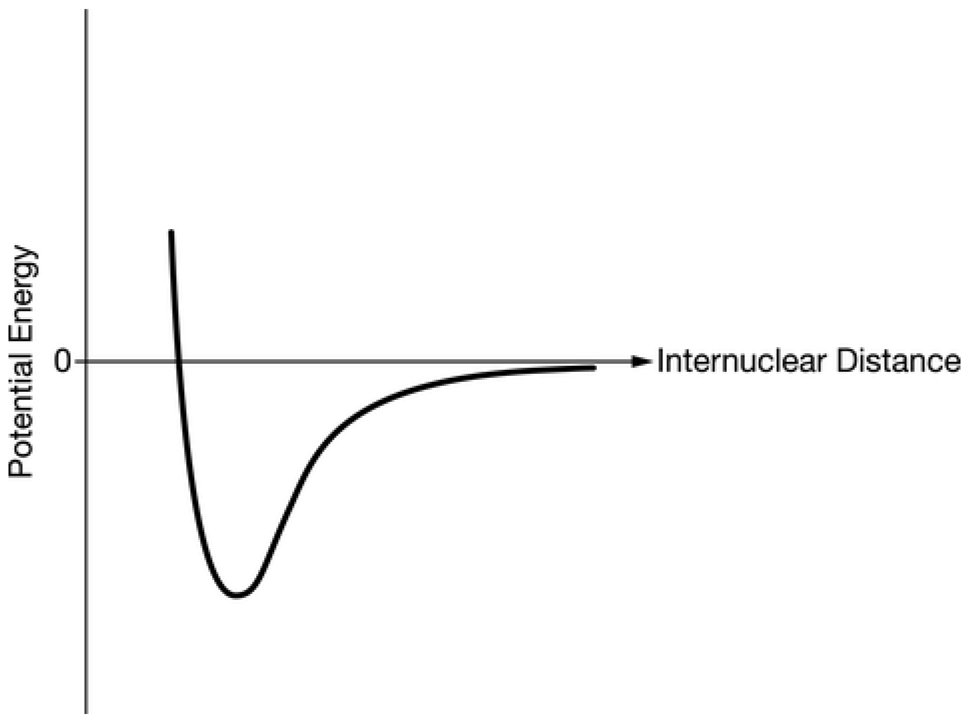
56. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to the chemical bonding in substances containing Cl.

(a) What type of chemical bond is present in the Cl_2 molecule?

(b) Cl_2 reacts with the element Sr to form an ionic compound. Based on periodic properties, identify a molecule, X_2, that is likely to to react with Sr in a way similar to how Cl_2 reacts with Sr. Justify your choice.

(c) A graph of potential energy versus internuclear distance for two Cl atoms is given below. On the same graph, carefully sketch a curve that corresponds to potential energy versus internuclear distance for two Br atoms.



(d) In the box below, draw a complete Lewis electron-dot diagram for the C_2Cl_4 molecule.

(e) Answer the following based on the diagram you drew above.

(i) What is the hybridization of the C atoms in C_2Cl_4 ?

(ii) What is the approximate chlorine-carbon-chlorine bond angle in C_2Cl_4 ?

(iii) Is the C_2Cl_4 molecule polar?

57. Answer the following questions using principles of chemical bonding and molecular structure.

(a) Consider the carbon dioxide molecule, CO_2 , and the carbonate ion, CO_3^{2-}.

(i) Draw the complete Lewis electron-dot structure for each species.

(ii) Account for the fact that the carbon-oxygen bond length in CO_3^{2-} is greater than the carbon-oxygen bond length in CO_2.

(b) Consider the molecules CF_4, and SF_4 .

(i) Draw the complete Lewis electron-dot structure for each molecule.

(ii) In terms of molecular geometry, account for the fact that the CF_4 molecule is nonpolar, whereas the SF_4 molecule is polar.

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58.

Nonmetal	C	N	O	Ne	Si	P	S	Ar
Formula of Compound	CF_4	NF_3	OF_2	No compound	SiF_4	PF_3	SF_2	No compound

Some binary compounds that form between fluorine and various nonmetals are listed in the table above. A student examines the data in the table and poses the following hypothesis: the number of F atoms that will bond to a nonmetal is always equal to 8 minus the number of valence electrons in the nonmetal atom.

a. Based on the student's hypothesis, what should be the formula of the compound that forms between chlorine and fluorine?

b. In an attempt to verify the hypothesis, the student researches the fluoride compounds of the other halogens and finds the formula ClF_3. In the box below, draw a complete Lewis electron-dot diagram for a molecule
of ClF_3.

c. Two possible geometric shapes for the ClF_3 molecule are trigonal planar and T-shaped. The student does some research and learns that the molecule has a dipole moment. Which of the two shapes is consistent with the fact that the ClF_3 molecule has a dipole moment? Justify your answer in terms of bond polarity and molecular structure.

In an attempt to resolve the existence of the ClF_3 molecule with the hypothesis stated above, the student researches the compounds that form between halogens and fluorine, and assembles the following list.

Halogen	Formula(s)
F	F_2
Cl	
Br	BrF, BrF_3, BrF_5
I	IF, IF_3, IF_5, IF_7

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d. Based on concepts of atomic structure and periodicity, propose a modification to the student's previous hypothesis to account for the compounds that form between halogens and fluorine.

59. Using principles of atomic and molecular structure and the information in the table below, answer the following questions about atomic fluorine, oxygen, and xenon, as well as some of their compounds.

Atom	First Ionization Energy ($kJ\ mol^{-1}$)
F	1,681.0
O	1,313.9
Xe	?

(a) Write the equation for the ionization of atomic fluorine that requires 1,681.0 $kJ\ mol^{-1}$.

(b) Account for the fact that the first ionization energy of atomic fluorine is greater than that of atomic oxygen. (You must discuss both atoms in your response.)

(c) Predict whether the first ionization energy of atomic xenon is greater than, less than, or equal to the first ionization energy of atomic fluorine. Justify your prediction.

(d) Xenon can react with oxygen and fluorine to form compounds such as XeO_3 and XeF_4 . In the boxes provided, draw the complete Lewis electron-dot diagram for each of the molecules represented below.

XeO_3 XeF_4

(e) On the basis of the Lewis electron-dot diagrams you drew for part (d), predict the following:

(i) The geometric shape of the XeO_3 molecule
(ii) The hybridization of the valence orbitals of xenon in XeF_4

(f) Predict whether the XeO_3 molecule is polar or nonpolar. Justify your prediction.